

Yichen Han

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Trustworthy and interpretable statistical learning for genomic biomarker discovery

Education

M.Sc. Statistics & Data Science

Sep 2027 (expected)

Ludwig-Maximilians-Universität München

Machine Learning track

B.Sc. Statistics & Data Science

Oct 2022 – Jan 2026

Ludwig-Maximilians-Universität München

Grade: 1.1 (German scale; 1 = best)

Minor in Biology (30 ECTS)

Thesis: *Robustness-Driven Feature Attribution in Genomic Deep Learning with Integrated Gradients*

Investigated when adversarial robustness improves interpretation plausibility in genomic CNNs. Supervised by Bernd Bischl and Martin Binder. Grade: 1.0.

[Abstract](#) · [Slides](#) · [GitHub](#)

Publications

Münch P.C., Safaei N., Mreches R., Binder M., **Han Y.**, Robertson G., Franzosa E.A., Huttenhower C.*, McHardy A.C.* (2025). *Comparative Assessment of Large Language Models for Microbial Phenotype Annotation*. Under review at *Genome Biology*. [bioRxiv](#).

(Contribution: statistical analysis, ensemble learning methods)

Aleksic M., Ehring T., Kunze A., **Han Y.**, Funk H., Wolkenstein L.* (2025). *Selective Effects of EMDR, Imagery Rescripting and Imaginal Exposure on Voluntary and Involuntary Memory of an Aversive Autobiographical Event*. R&R at *Journal of Experimental Psychology: Applied*. [OSF](#).

(Contribution: statistical guidance, methods & results review)

Lin L., Hatzikotoulas K., ..., Wiegrebbe S., **Han Y.**, Ivanova O., ..., Zeggini E., Rachow A.*, Held K.* (2026). *Genetic Insights into Lung Function and Its Trajectory among People with Tuberculosis in Africa: A Genome-Wide Association Study*. Under review at *European Respiratory Journal*.

(Contribution: statistical guidance, methods & results review)

*Senior/corresponding author

Research Experience

The Huttenhower Lab, Harvard T.H. Chan School of Public Health

Jan 2025 – present

Collaborating Student Researcher (joint with Philipp Münch)

Boston, MA, USA (remote)

- Leading the interpretable genomic deep learning project, advised by Curtis Huttenhower
- Designed synthetic genome datasets for interpretability benchmarking, identifying compositional biases as key confounders
- Developed evaluation framework for assessing biological plausibility of model explanations
- Presented project findings at the departmental research group seminar (Aug 2025)

BIFO Group, Helmholtz Centre for Infection Research

Oct 2024 – present

Research Assistant (PI: Alice McHardy; Team lead: Philipp Münch)

Braunschweig, Germany (remote)

- Developed adversarial training pipelines for genomic CNNs (PyTorch)
- Implemented Integrated Gradients for genomic models (Python, R)
- Developed statistical tools for Rashomon effect evaluation, revealing interpretation instability in real data

Statistical Consulting Unit (StaBLab), LMU Munich

Research Assistant & Statistical Consultant

Apr 2024 – present

Munich, Germany

- Methodological support to 30+ graduate projects (MSc/PhD) in biology, medicine, psychology
- Methods: GLMMs/GAMs, Bayesian methods, imputation, spatial-temporal, causal DAGs, applied DL
- Contributing to BfS-funded Wismut project: Bayesian hierarchical modeling of exposure uncertainties in radiation risk assessment; HPC pipeline development

Academic Projects (selected)

The Misspecification Bias in TSLS with Covariates

Oct 2024 – Mar 2025

Causal Inference Seminar (Master's level), Grade: 1.0, [GitHub](#)

LMU Munich

- Supervised by Daniel Wilhelm.
- Investigated conditions under which Two-Stage Least Squares (TSLS) with covariates yields valid causal interpretations, synthesizing recent results on “rich covariates” for positively weighted treatment effects
- Demonstrated misspecification bias through simulations; surveyed uptake in subsequent literature

Wildlife abundance modeling

Feb 2025 – May 2025

Advanced Statistical Practicum, Grade: 1.0

LMU Munich

- Evaluated N-Mixture models for estimating wildlife abundance from repeated camera-trap data across ten German national parks (collaboration with Nationalparkverwaltung Bayerischer Wald)
- Wrote a review on N-Mixture model applications and theoretical gaps

Teaching Experience (selected)

Teaching Assistant

Oct 2023 – Mar 2026

Department of Statistics

LMU Munich

- **Tutorial Instructor** – Descriptive Statistics & EDA (Winter 2023/24): weekly tutorials for ~120 students; designed exercises
- **Project Supervisor** – Advanced Statistical Practicum (Winter 2025/26): supervised 2 student projects (Greek tomb kinship modeling; stroke metabolomics)
- **Curriculum Developer** – Descriptive Statistics & Probability Theory (Summer 2024): primary author of exam-prep materials; workshops for ~80 students

Service & Leadership

Elected Member, Faculty Council (*Fakultätskonvent*)

Jul 2025 – Jul 2026

Faculty of Mathematics, Informatics and Statistics

LMU Munich

Student Representative

Mar 2025 – present

Two W2 Professorship Appointment Committees

Department of Statistics, LMU Munich

Student Council Statistics & Data Science

Apr 2023 – present

Elected Member (Jul 2025 – Jul 2026); Lead, Teaching Working Group

LMU Munich

Skills

Programming R · Python · SAS · Bash

Tools SLURM, Snakemake, Docker, SQL, Git, Linux, $\text{\LaTeX}$, Inkscape

Languages Chinese (native), German (C1), English (IELTS 8.0)